



Department of Computer Science & Engineering

Ajay Kumar Garg Engineering, Ghaziabad

Final Year Project Title and Abstract

Submit it before 26 August 2025

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Project Title (should not be a one/two words)	A Hybrid Data-Driven and Distributionally Robust Optimization Framework for Optimal Sizing of PV–BESS Microgrids in Uncertain and Data-Scarce Environments
Project Description (100 words)	Reliable electricity access in remote regions remains a major challenge, with renewable microgrids offering a practical solution. This project proposes a data-driven optimal sizing framework for distributed energy resources (DERs), integrating photovoltaic (PV) generation and battery storage. Using hybrid physical modeling and machine learning, it ensures accurate and efficient design under uncertain demand and variable resources. To address noisy or scarce datasets, distributionally robust optimization (DRO) enhances resilience while reducing over/under-sizing risks. The framework reduces design time, improves cost-effectiveness, and supports scalable, reliable microgrid deployment in energy-deficient regions.
Programming Languages	Python, Streamlit, Flask (coding), Tableau, PowerBi (visualization)
Project Mentor Name and Sign	Dr. Ashish Dixit
Project Coordinator Name and sign	Dr. Ashish Dixit
PI Name and Sign	Dr. Avdhesh Gupta